1 Variation of parameters
- Announcements
- Variation of parameters
- Recap
- Second-order equations
- Higher-order equations
- Practice problems

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 20
Variation of parameters
Thursday 29 October 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Variation of Parameters

[Trench §5.7], [Trench §9.4].

Recap

So far, we have learned how to

- Solve $L[y] = 0$ if L has constant coefficients
  - Try $y = e^{rx} \implies$ characteristic equation for r
- Solve $L[y] = g(x)$ if L has constant coefficients and $g(x)$ can be written as sums and products of constants, exponentials, sines and cosines
  - Solve $L[y] = 0$ for *complementary function* y_c
  - Use method of undetermined coefficients to find *particular solution* y_p
  - General solution is $y = y_c + y_p$

What if L does not have constant coefficients? or $g(x)$ is not of the form that is amenable to the method of undetermined coefficients?

Solving $L[y] = g(x)$ for any $g(x)$

Consider the ***general 2nd order linear ODE***

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x) \tag{1}$$

where $a_i(x)$ and $g(x)$ are continuous on an interval $\mathcal{I}$ and $a_2(x) \neq 0$ on $\mathcal{I}$.

We can write Equation (1) in ***standard form***,

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \tag{2}$$

where

$$P(x) = \frac{a_1(x)}{a_2(x)}, \quad Q(x) = \frac{a_0(x)}{a_2(x)}, \quad \text{and} \quad f(x) = \frac{g(x)}{a_2(x)}$$

are continuous on $\mathcal{I}$.

Note: If P and Q are *constant*, then we can always find the complementary function y_c, *i.e.*, the general solution to $L[y] = y'' + Py' + Qy = 0$.

Solving $L[y] = g(x)$ for any $g(x)$

Consider the general 2nd order linear ODE in standard form,

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \qquad (2)$$

- If $P = \text{const}$ and $Q = \text{const}$, we can always solve $L[y] = 0$.
- If P and/or Q are not constant, we cannot necessarily find the general solution to $L[y] = 0$.
- For some non-constant functions $P(x)$ and $Q(x)$, we can solve $L[y] = 0$ (we'll see some examples later).
- For now, we'll assume we have solved $L[y] = 0$, so we have a fundamental solution set $\{y_1, y_2\}$ for $L[y] = 0$, and hence a complementary function $y_c(x) = c_1 y_1(x) + c_2 y_2(x)$ for $L[y] = f(x)$.

Can we find a particular solution y_p for $L[y] = f(x)$?

Method of variation of parameters

Earlier, we used the idea of ***variation of parameters*** to derive the integrating factor for a first order linear ODE.

We now use the *same idea* to derive a *particular solution* y_p for a *second order* linear ODE,

$$L[y] \;=\; y'' + P(x)y' + Q(x)y \;=\; f(x) \qquad (2)$$

for which we know the *complementary function*,

$$y_c(x) = c_1 y_1(x) + c_2 y_2(x), \qquad c_1, c_2 \in \mathbb{R}.$$

Our *stroke of genius* is to guess that y_p can be expressed similarly to y_c if we allow the parameters, c_1 and c_2, to vary, *i.e.*, if we replace c_1 and c_2 with functions $u_1(x)$ and $u_2(x)$.

Method of variation of parameters

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \qquad (2)$$

So suppose that $L[y_1] = 0$ and $L[y_2] = 0$, and let

$$y_p = u_1 y_1 + u_2 y_2 \qquad \textit{and then solve for } u_1(x) \textit{ and } u_2(x)$$

$$\begin{aligned}
\implies y_p' &= \frac{\mathrm{d}}{\mathrm{d}x}\Big(u_1 y_1 + u_2 y_2\Big) \\
&= u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2' \\
\implies y_p'' &= \frac{\mathrm{d}}{\mathrm{d}x}\Big(u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'\Big) \\
&= u_1'' y_1 + u_1' y_1' + u_1' y_1' + u_1 y_1'' + u_2'' y_2 + u_2' y_2' + u_2' y_2' + u_2 y_2''
\end{aligned}$$

We now write out $L[y_p]$ and simplify using the fact that $L[y_1] = 0$ and $L[y_2] = 0$. We will choose u_1 and u_2 so that $L[y_p] = f(x)$.

Method of variation of parameters

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \qquad (2)$$

Assuming y_p is a particular solution, *i.e.*, $f(x) = L[y_p]$, we have

$$\begin{aligned} f(x) &= y_p'' + P y_p' + Q y_p \\ &= u_1 L[y_1] + u_2 L[y_2] + \text{other terms} \\ &\vdots \\ &= \frac{\mathrm{d}}{\mathrm{d}x}\left(y_1 u_1' + y_2 u_2'\right) + P\left(y_1 u_1' + y_2 u_2'\right) + y_1' u_1' + y_2' u_2' \end{aligned}$$

This equation will be satisfied if we insist that

$$\begin{aligned} & y_1 u_1' + y_2 u_2' = 0 \\ \text{and} \quad & y_1' u_1' + y_2' u_2' = f(x) \end{aligned}$$

This is a system of two linear (algebraic) equations for $\{u_1', u_2'\}$.

Method of variation of parameters

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \tag{2}$$

We have found that if $\{y_1, y_2\}$ is a fundamental solution set for $L[y] = 0$, and we define $y_p = u_1 y_1 + u_2 y_2$ then

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ f(x) \end{pmatrix} \implies L[y_p] = f(x)$$

We can solve the linear system for $\{u_1', u_2'\}$ using any method we like, but it is convenient to use ***Cramer's Rule***,

$$A\vec{v} = \vec{b} \implies v_i = \frac{\det A_i}{\det A}$$

where A_i is the matrix obtained by replacing the ith column of A with $\vec{b}$.

Note: If $A = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}$ then $\det A = W(y_1, y_2)$, the Wronskian.

Method of variation of parameters

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \qquad (2)$$

Writing $A = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}$, A_i = the matrix obtained by replacing the ith column of A with the column vector $(0, f(x))^{\mathsf{T}}$, and $W_i = \det(A_i)$, we have

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \qquad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \qquad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix},$$

and hence, from Cramer's Rule,

$$u_1' = \frac{W_1}{W} = -\frac{y_2 f(x)}{W}, \qquad u_2' = \frac{W_2}{W} = \frac{y_1 f(x)}{W}$$

which are well defined because W is the Wronskian of a fundamental solution set, so $W \neq 0$ on the interval $\mathcal{I}$.

Method of variation of parameters

$$L[y] = y'' + P(x)y' + Q(x)y = f(x) \qquad (2)$$

Thus, given a complementary function $y_c = c_1 y_1 + c_2 y_2$, a particular solution of Equation (2) is

$$y_p = u_1 y_1 + u_2 y_2$$

where

$$u_1 = \int \frac{W_1}{W}\,dx = -\int \frac{y_2 f(x)}{W}\,dx\,, \qquad u_2 = \int \frac{W_2}{W}\,dx = \int \frac{y_1 f(x)}{W}\,dx$$

Note: We can ignore arbitrary constants in the antiderivatives above because $L[C\,y_i] = C\,L[y_i] = 0$ for constant C.

Note: Our W_i denotes the full column-replacement determinant in Cramer's rule. Trench [§9.4, Eq. (9.4.6)] instead uses W_j for the determinant obtained by deleting the last row and column j of the Wronskian matrix, writing the cofactor sign and forcing factor separately. Trench's formulae give the same u_i'.

Method of variation of parameters

Example ($y'' + y = \sec(x), \quad -\pi/2 < x < \pi/2$)

First solve homogeneous equation, $y'' + y = 0$.

Characteristic equation: $r^2 + 1 = 0 \implies r^2 - i^2 = 0 \implies$
$(r - i)(r + i) = 0 \implies e^{ix}, e^{-ix}$ are complex solutions $\implies$
$\cos x, \sin x$ are real solutions $\implies y_c = c_1 \cos x + c_2 \sin x$.

Please do poll *Choosing a method for secant forcing* on childsmath...

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = -\sin x \sec x = -\tan x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = \cos x \sec x = 1$$

Method of variation of parameters

Example ($y'' + y = \sec(x), \quad -\pi/2 < x < \pi/2$ (continued))

Therefore,

$$\begin{aligned} u_1(x) &= \int \frac{W_1}{W}\,dx = \int \frac{-\tan x}{1}\,dx = -\int \tan x\,dx \\ &= -\ln(\sec x) = \ln(\cos x) \\ u_2(x) &= \int \frac{W_2}{W}\,dx = \int \frac{1}{1}\,dx = \int dx = x \end{aligned}$$

Therefore,

$$y_p = u_1 y_1 + u_2 y_2 = \ln(\cos x)\cos x + x \sin x$$

and hence

$$y = y_c + y_p = [c_1 + \ln(\cos x)]\cos x + [c_2 + x]\sin x$$

Method of variation of parameters for n^{th} order ODEs

The nth order linear ODE in *standard form* is

$$L[y] = y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y' + P_0(x)y = f(x)$$

where $P_i = a_i/a_n$ $(i = 0, \ldots, n-1)$ and $f = g/a_n$.
If we can find a fundamental solution set, $\{y_1, \ldots, y_n\}$, for the associated homogeneous equation, $L[y] = 0$, then we can use *variation of parameters* to find a particular solution

$$y_p = u_1 y_1 + u_2 y_2 + \cdots + u_n y_n,$$

with

$$u_i(x) = \int \frac{W_i}{W}\, \mathrm{d}x\,,$$

where W is the Wronskian and W_i is the determinant obtained by replacing the ith column in the Wronskian with $(0, \ldots, 0, f(x))^{\mathsf{T}}$.

Note: Trench [§9.4, Eq. (9.4.1)] uses P_j for our a_{n-j} and F for our g, before normalization. Our coefficient subscripts match the derivative order.

Method of variation of parameters for n^{th} order ODEs

Example ($y''' - y'' + y' - y = \sec x, \quad -\pi/2 < x < \pi/2$)

characteristic equation: $r^3 - r^2 + r - 1 = (r-1)(r^2+1) = (r-1)(r-i)(r+i) \implies y_c = c_1 e^x + c_2 \cos x + c_3 \sin x$

Therefore,

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^x & \cos x & \sin x \\ e^x & -\sin x & \cos x \\ e^x & -\cos x & -\sin x \end{vmatrix} = 2e^x$$

$$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ \sec x & -\cos x & -\sin x \end{vmatrix} = \sec x(\cos^2 x + \sin^2 x) = \sec x$$

$$\implies u_1(x) = \int \frac{W_1}{W}\,dx = \int \frac{\sec x}{2e^x}\,dx = \frac{1}{2}\int e^{-x}\sec x\,dx$$

Method of variation of parameters for n^{th} order ODEs

Example ($y''' - y'' + y' - y = \sec x, \quad -\pi/2 < x < \pi/2$ (continued.))

$$W_2 = \begin{vmatrix} e^x & 0 & \sin x \\ e^x & 0 & \cos x \\ e^x & \sec x & -\sin x \end{vmatrix}$$

$$= -\sec x(e^x \cos x - e^x \sin x) = e^x(\tan x - 1)$$

$$\Longrightarrow \quad u_2(x) = \int \frac{W_2}{W}\,dx = \int \frac{e^x(\tan x - 1)}{2e^x}\,dx$$

$$= \frac{1}{2}\int (\tan x - 1)\,dx = \frac{1}{2}(\ln(\sec x) - x)$$

Method of variation of parameters for n^{th} order ODEs

Example ($y''' - y'' + y' - y = \sec x, \quad -\pi/2 < x < \pi/2$ (continued..))

$$W_3 = \begin{vmatrix} e^x & \cos x & 0 \\ e^x & -\sin x & 0 \\ e^x & -\cos x & \sec x \end{vmatrix}$$

$$= \sec x(-e^x \sin x - e^x \cos x) = -e^x(\tan x + 1)$$

$$\Longrightarrow \quad u_3(x) = \int \frac{W_3}{W}\,dx = \int \frac{-e^x(\tan x + 1)}{2e^x}\,dx$$

$$= -\frac{1}{2}\int (\tan x + 1)\,dx = -\frac{1}{2}(\ln(\sec x) + x)$$

Method of variation of parameters for n^{th} order ODEs

Example ($y''' - y'' + y' - y = \sec x, \quad -\pi/2 < x < \pi/2$ (continued...))

Putting together our calculations we have

$$\begin{aligned} y_c &= c_1 y_1 + c_2 y_2 + c_3 y_3 \qquad\qquad c_1, c_2, c_3 \in \mathbb{R} \\ &= c_1 e^x + c_2 \cos x + c_3 \sin x \\ y_p &= u_1 e^x + u_2 \cos x + u_3 \sin x \\ &= \left(\frac{1}{2}\int e^{-x} \sec x \, dx\right) e^x + \left(\frac{1}{2}\big(\ln(\sec x) - x\big)\right) \cos x \\ &\qquad - \left(\frac{1}{2}\big(\ln(\sec x) + x\big)\right) \sin x \end{aligned}$$

How do we evaluate $\int e^{-x} \sec x \, dx$?

Express it either via special functions (not in this course) or write it as a definite integral that we can evaluate numerically:

$$\int e^{-x} \sec x \, dx = \int_0^x e^{-t} \sec t \, dt$$

Method of variation of parameters for n^{th} order ODEs

Example ($y''' - y'' + y' - y = \sec x, \quad -\pi/2 < x < \pi/2$ (continued....))

So, finally, the *general solution* for this example is

$$\begin{aligned} y &= y_c + y_p \\ &= c_1 e^x + c_2 \cos x + c_3 \sin x \\ &\quad + \frac{1}{2} e^x \int_0^x e^{-t} \sec t \, \mathrm{d}t \\ &\quad + \frac{1}{2} \cos x \big(\ln(\sec x) - x \big) \\ &\quad - \frac{1}{2} \sin x \big(\ln(\sec x) + x \big) \end{aligned}$$

where $c_1, c_2, c_3 \in \mathbb{R}$. □

Practice problems

- Second-order variation of parameters [Trench §5.7]:
  - Exercises 1–6: constant-coefficient equations.
  - Exercises 7, 8, 11, 13, 17, 22: a fundamental set is supplied.
  - Exercises 30–32: initial value problems.
- Higher-order variation of parameters [Trench §9.4]:
  - Exercises 3, 11, 12: particular solutions.
  - Exercises 22, 23: initial value problems.